

VISHNU PURUSHOTHAMAN

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EDUCATION

The University of Texas at Austin

Expected May 2029

Bachelor of Science in Computer Science | GPA: 3.93 / 4.00

Austin, TX

Relevant Coursework: Data Structures, Discrete Mathematics, Computer Organization & Architecture, Linear Algebra, Statistics & Data Science, Multivariable Calculus

PROJECTS

SnapNotes — iOS piano-learning app (published) | [App Store](#) | [Website](#)

Summer 2026 – Present

Flutter/Dart, Swift, Python/FastAPI, Docker, Fly.io, Audiveris OMR, StoreKit, CoreMIDI, AVFoundation

- Designed, built, and shipped a solo iOS/iPadOS app (1k+ downloads) that converts photographed or PDF sheet music into an interactive falling-notes practice experience; monetized with a free tier and Pro subscription via StoreKit
- Architected a Flutter front end (BLoC state management, custom-drawn 60 fps note visualizer, MIDI keyboard input over USB/Bluetooth) with native Swift integrations for document scanning, home-screen widgets, and iCloud backup
- Deployed an optical music recognition backend (FastAPI, Audiveris, Tesseract OCR) in Docker on Fly.io with token auth, scale-to-zero, and result caching; maintained a 254-song catalog and ~370 automated tests through TestFlight and App Store release

Moontide — iOS/iPadOS arcade game (published) | [App Store](#) | [Website](#)

Sep 2026

Swift, SwiftUI, Metal, CoreHaptics, AVAudioEngine, StoreKit 2, Game Center, iCloud, XCTest

- Built a one-touch physics game solo, prototyped on the web then ported to native Swift with zero dependencies; golden-trace tests replay 1,100 recorded frames to verify parity with the original to within 1e-11 px
- Wrote a custom Metal water shader and velocity-driven CoreHaptics on a 120 Hz render loop; split the game into Core/Rendering/Audio/Platform packages with an event-driven core covered by 130+ tests; shipped 4 updates in 2 weeks

Companion: Diabetes Journal — iOS + watchOS health app (submitted to App Store)

2026

Swift 6, SwiftUI, SwiftData, HealthKit, CloudKit, Foundation Models, StoreKit 2, WidgetKit, App Intents

- Built a local-first diabetes journal solo under Swift 6 strict concurrency: iPhone app, Apple Watch app with complications, Lock Screen widgets, Live Activities, Control Center controls, Siri App Intents, and CloudKit sync
- Shipped an on-device conversational assistant on Apple's Foundation Models framework with a tool-grounded session (10 data tools), streaming replies, and deterministic routing gates so the model can only cite real logged data
- Designed a deterministic insight engine and glucose-analytics layer (ambulatory glucose profile, variability, food response) verified by a 127-test Swift Testing suite; owned StoreKit 2 subscriptions and a CLI archive-and-upload pipeline (16 builds)

Talk Up — speech-support iOS app

2025

- Co-developed a Swift/UIKit app using CoreML to give users with speech disabilities real-time feedback on clarity, tone, and pacing

Self-Driving Car Simulation — browser neural network, zero dependencies

2025

- Implemented car physics, ray-cast sensors, and a feed-forward neural network with evolutionary training in pure JavaScript

EXPERIENCE

Android App Development Intern — The University of Texas at Dallas, Dept. of Computer Science

Jun – Aug 2024

- Developed multiple Android apps in Java/XML, culminating in a multithreaded game with an enhanced Pong ruleset; recognized by Prof. John Cole for rapid mastery of advanced concepts

CS Entrepreneurship Intern — Science Gurus (Remote)

Jun – Jul 2024

- Completed a 6-week intensive on product strategy, market research, UX design, prototyping, and go-to-market planning

LEADERSHIP & ACTIVITIES

Texas Game Jam — Participant

Sep 2025, Sep 2026

- Built a solo 2D game in Unity/C# in 48 hours at a statewide event judged by industry professionals
- 2026: co-built [Return to Sender](#) in Godot, a parry-based bullet-hell with 3 multi-phase bosses, 4 modes, 39 voiced lines, and full accessibility options

Flower Mound HS Computer Science Club — Officer

2022 – 2025

- Organized hackathons and mentored peers; placed 6th (team) in the HPE CodeWars Novice Division

Flower Mound HS Symphonic Orchestra — Concertmaster Violinist

2021 – 2025

- Led a 50+ member orchestra as concertmaster, running rehearsals and coordinating section leaders

SKILLS

Languages: Swift, Dart, Java, Python, JavaScript/HTML/CSS, C#, SQL

Mobile & Frameworks: SwiftUI, UIKit, SwiftData, HealthKit, CloudKit, WidgetKit, App Intents, StoreKit 2, CoreML, CoreMIDI, AVFoundation, Metal, CoreHaptics, Foundation Models, Flutter (BLoC), Android (Java/XML), Unity, Godot, FastAPI

Tools & Infrastructure: Git, Docker, Fly.io, Xcode, App Store Connect/TestFlight, Swift Testing, XCTest, Audiveris, Tesseract, music21

Certifications: Python for Everybody (University of Michigan); Mathematics for Machine Learning (Imperial College London)